Harish G

Agentic AI & Automation Engineer | LLM Guardrails, Agent Orchestration, RAG, Document Intelligence

Bengaluru, India | +91 7892855850 | harishgreddy.work@gmail.com | LinkedIn | GitHub | harishg.com

Profile

I build LLM systems that cannot invent a number. The pattern is the same in everything I ship: deterministic code holds authority, the model is a constrained component, and any figure it states is checked against the tool result that produced it. That means numeric guardrails that rewrite unverified output, retrieval that runs before the model with role filters inside the retriever, extraction that returns null rather than guess, and human approval gates an agent cannot talk its way past. Software Developer at CortexCraft.ai, in Python/FastAPI and Java/Spring Boot.

Core Skills

AI Guardrails & Control: numeric provenance guardrails, retrieval-before-inference, deterministic-first fallback chains, refusal-by-design parsers, human approval gates and safe escalation, role-scoped retrieval, append-only audit trails, frozen evaluation fixtures

Agentic AI: LLM function and tool calling (runtime Gemini function declarations, bidirectional tool-call loop), config-driven tool registries, MCP-style tool hosting (`inputSchema`, `readOnlyHint`, `destructiveHint`), prompt frameworks

RAG & Retrieval: pgvector (HNSW, cosine distance), Gemini `text-embedding-004` and MiniLM embeddings, chunking with overlap, Postgres full-text fallback, role-scoped row-level retrieval, Neo4j graph-enriched retrieval

Document Intelligence: PDF / DOCX / XLSX / CSV extraction with magic-byte validation, schema-driven LLM field extraction with per-type coercion, closed-label classification, summarisation with sentiment and aspect tagging, on-device OCR (ML Kit)

Process Automation & Integration: throttled job runners, guarded state machines, do-not-call screening, retry and backoff policy, Celery and APScheduler queues, CSV / Excel pipelines, desktop UI automation; Twilio, Plivo, WhatsApp Business API, Google Calendar, Jira, GitHub, REST and webhook APIs

Languages, Backend & Delivery: Python, Java, JavaScript / TypeScript, Kotlin, SQL; FastAPI, Spring Boot, Node / Express, Celery, Redis; PostgreSQL / pgvector, MongoDB, Neo4j; Docker, GitLab CI, GitHub Actions, Jenkins, pytest with coverage gates

Experience

Software Developer, CortexCraft.ai Jan 2026 – Present

- Authored 380 of 418 commits on the voice-agent SaaS platform: agent orchestration, campaign automation, transcript intelligence and tenant governance, behind 92 pytest suites, Playwright end-to-end tests and a GitLab CI pipeline.
- Built human-in-the-loop escalation along two independent paths, so the guarantee survives the model failing: a `flag_human_request` tool the agent calls, and a transcript-pattern safety net that sets the same state when it does not. Live escalation dials a person into the running call through a Twilio conference.
- Built transcript document intelligence on Gemini JSON mode: schema-driven field extraction that coerces each value to its declared type and nulls anything off-type rather than fabricating it, plus summarisation with key points, sentiment and per-topic aspect tags.
- Built the agent tool layer: runtime Gemini function declarations from agent config, a bidirectional tool-call loop and a generic HTTP tool executor, so onboarding a customer system is configuration rather than code.
- Built the campaign automation runner: tick-driven dispatch throttled by concurrency and calls-per-hour, do-not-call screening before dial, retry with configurable delay and attempt limits, behind a guarded state machine that rejects invalid transitions with HTTP 409.

Freelance Web & App Developer (Self-Employed) 2023 – 2026

Delivered full-stack web and app projects end to end for clients: React front ends, Python and Node services, and automation scripts.

Selected Projects

AI Personal CFO and Talo *Kotlin, Jetpack Compose, Hilt, Room/SQLCipher, ML Kit, Gemini Live*

- Sole author, 159 commits, 34 Gradle modules, 199 test files. Twelve deterministic engines compute every figure and the model is only permitted to explain them, on a written rule: numbers from math, words from AI.
- An output guardrail rewrites any number in model text that is absent from the tool result as `[unverified]`. An MCP-style in-process tool host exposes the engines through a JSON registry carrying `inputSchema`, `readOnlyHint` and `destructiveHint`; chat reads through the registry and never writes.
- On-device ML Kit OCR extracts receipt fields; the SMS parser is built to refuse, resolving every ambiguous case to null. Both are scored against frozen evaluation fixtures.

LeadCall AI and AI Code-Review Gate *FastAPI, Celery, PostgreSQL, React, Ollama, GitHub API*

- An approval gate for AI-generated code changes that an agent cannot argue past: high-risk changes are blocked from approval outright, a rejection requires a written reason and regenerates with that reason fed back, and every reviewer action lands in an append-only audit log. Nothing auto-merges.
- Multi-tenant outbound calling underneath it: Celery dial and callback workers, Redis dial locks, per-tenant telephony credentials encrypted at rest, and Excel lead import.

Dental Clinic AI Assistant *Python, Flask, PostgreSQL/pgvector, Gemini, WhatsApp Business API*

- Chunked document RAG (800-token chunks, 100 overlap) embedded with `text-embedding-004` into pgvector, retrieved by cosine distance with a Postgres full-text fallback and role-scoped row filtering applied inside the retriever, so authorisation cannot be prompted around.
- Gemini function calling for booking with preconditions encoded in the tool descriptions, a role-based tool router admitting one tool per turn, and a 19-state WhatsApp conversation state machine that hands off to a phone agent with context preserved.

KisanVoice AI *Python, FastAPI, MongoDB, WhatsApp Flows, Google STT/TTS, Jenkins, Docker*

- 172 of 192 commits. Multilingual voice surveys across 24 locales, with programmatic WhatsApp Flow form generation, bulk CSV invite jobs and Excel round-trip, over an append-only audit log with TTL retention and ownership guards.

Education & Ways of Working

B.E., Information Science and Engineering, Vivekananda Institute of Technology, Bangalore. One issue per branch, decisions recorded as ADRs before the code lands, tests as the merge gate.